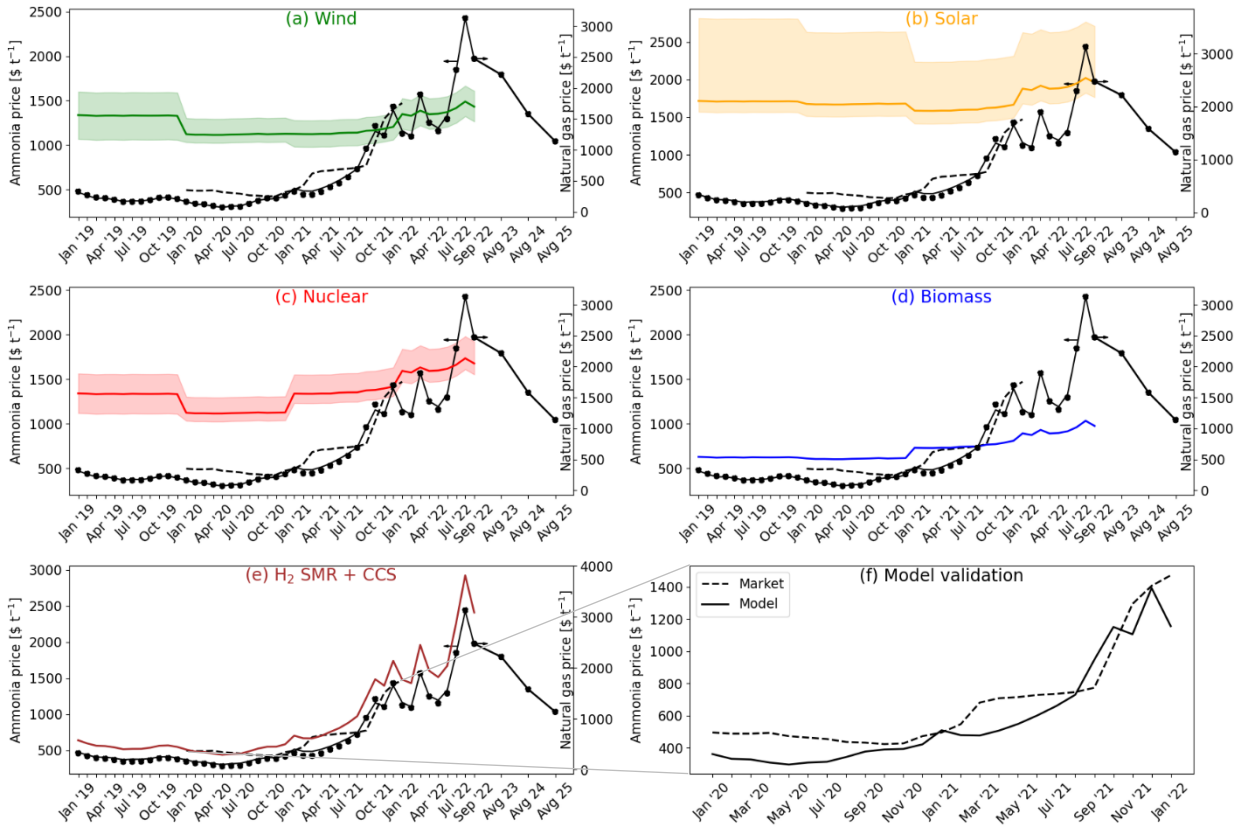

Fig. 1 BAU ammonia vs green ammonia with hydrogen from (a) water electrolysis using on-shore wind electricity, (b) water electrolysis using solar electricity, (c) water electrolysis using nuclear electricity, (d) biomass gasification, and (e) SMR + CCS. The dashed line in plots (a) – (e) represent the market price of ammonia. The shaded region in plots (a) – (c) is the sensitivity analysis due to the levelized cost of hydrogen (table 5). (f) shows the cost of production calculated using our model vs the market price. All the data points are shown in table 8.

- Since July 2021, natural gas prices have increased sharply (table 1). Similarly, the price of grid electricity has almost quadrupled (table 1). The major reasons for this trend are the aftermath of COVID-19, soaring inflation, climate change, and war on Ukraine.
- In the case of BAU ammonia, every ton of ammonia produced consumes 0.53 tons of natural gas and 940 kWh of electricity. As a consequence, the cost of production of BAU ammonia has increased many folds since July 2021. As of September 2022, cost of production for BAU ammonia is three times in comparison to that in July 2021.
- When natural gas prices reach around 1400$ t^{-1}, ammonia with hydrogen from water electrolysis using on-shore wind electricity becomes cost competitive with BAU ammonia. Also, at the peak of natural gas prices in August 2022, the wind scenario is 1.63 times lower than BAU ammonia.
- The solar and nuclear scenarios are only feasible around the peak of natural gas prices. For solar, this can be attributed to lower capacity factors (16.1% to 17.5%) in comparison to wind electricity (35.6% to 39%) resulting in higher levelized costs of hydrogen.

- The cost of construction and operation of nuclear power plants is becoming increasingly expensive and difficult. Therefore, even though they have a higher capacity factor of around 95%, the cost of production of electricity from nuclear energy is increasing.
- The biomass scenario is the best performing case out of all the alternatives. At natural gas prices of around 690$ t^{-1}, the biomass scenario is already economically more feasible than BAU ammonia. At the peak, BAU ammonia is 2.35 times more expensive than ammonia using hydrogen from biomass.
- Hydrogen from SMR + CCS is always more expensive than BAU ammonia due to the inclusion of CCS and also the additional nitrogen feed required in this scenario. Whereas, in BAU ammonia the nitrogen comes from air used in the autothermal reformer.

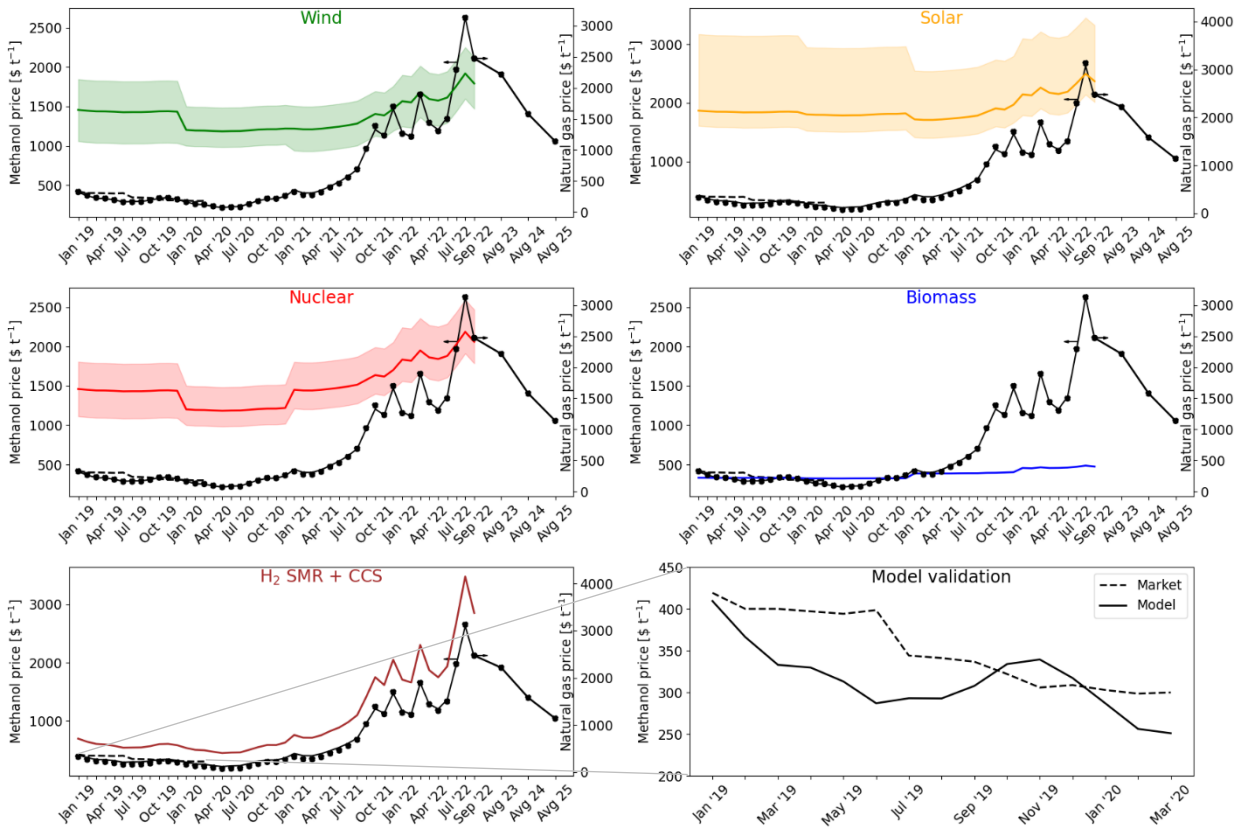

Fig. 2 BAU methanol vs green methanol with hydrogen from (a) water electrolysis using on-shore wind electricity, (b) water electrolysis using solar electricity, (c) water electrolysis using nuclear electricity, (d) biomass gasification, and (e) SMR + CCS. The dashed line in plots (a) – (e) represent the market price of methanol. The shaded region in plots (a) – (c) is the sensitivity analysis due to the levelized cost of hydrogen (table 5). (f) shows the cost of production calculated using our model vs the market price. All the data points are shown in table 9.

- In the case of BAU methanol, every ton of methanol produced consumes 0.67 tons of natural gas and 708 kWh of electricity. Therefore, similar to ammonia, the cost of production for BAU methanol has increased sharply. In comparison to July 2021, the cost of production of methanol rose by 243% in September 2022.

- For the cases of hydrogen from water electrolysis using (a) on-shore wind, (b) solar, (c) nuclear, and (e) SMR + CCS, carbon dioxide from Direct Air Capture (DAC) is used as a feed. For every ton of carbon dioxide captured, 0.11 ton of natural gas is required.
- At natural gas prices of around 1700\$ t^{-1}, the wind scenario is economically competitive in comparison to BAU methanol. Also, considering low prices of wind electricity, at natural gas prices of 1400\$ t^{-1}, methanol production with hydrogen from water electrolysis using on-shore wind electricity and carbon dioxide from DAC is more feasible.
- The solar and nuclear scenarios follow a similar trend to that seen for ammonia due to similar reasons as discussed before.
- The biomass scenario is again the best performing case out of all the alternatives considered. In this case, 1.74 tons of biomass (straw) and 100.5 kWh of electricity is required per ton of methanol produced. As of September 2022, production of BAU methanol is 4.4 times more than that of methanol from biomass.
- For the case of hydrogen from SMR + CCS, the cost of production is on average 67.8% higher than BAU methanol due to additional cost of CCS and carbon dioxide from DAC.

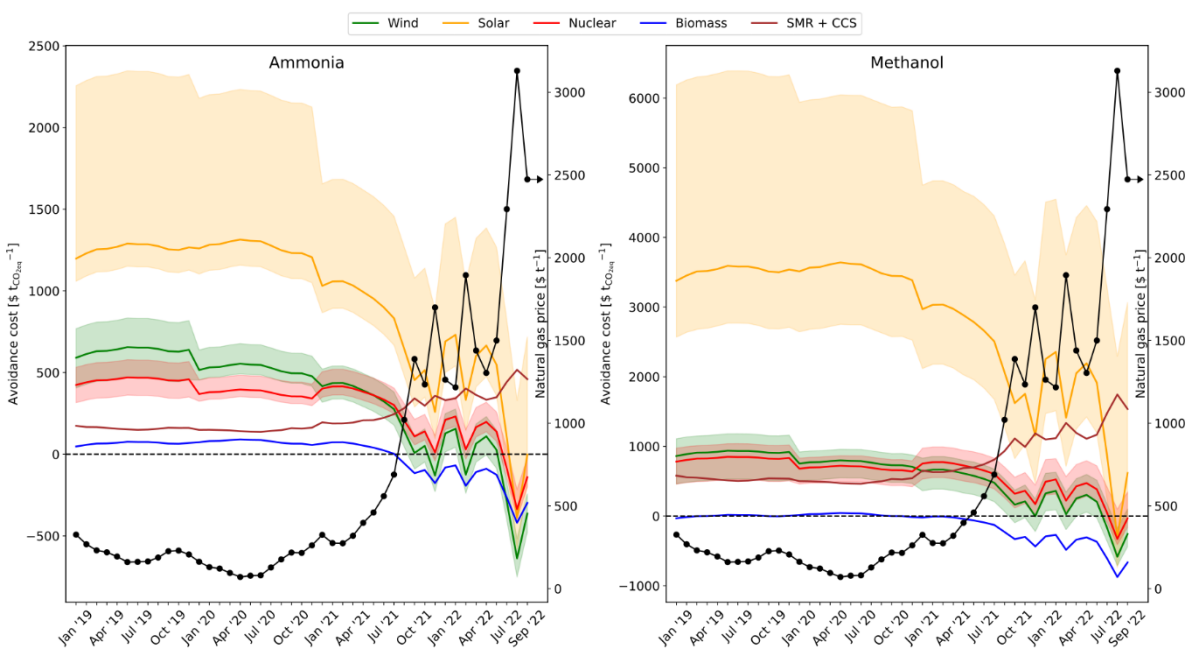

Fig. 3 CO_2 avoidance costs for (a) ammonia, and (b) methanol. The GWP values used are given in table 6 and table 7 for ammonia and methanol, respectively.

- CO_2 avoidance costs = $\frac{P.C_{alt} - P.C_{BAU}}{GWP_{BAU} - GWP_{alt}}$
  a) CO_2 avoidance costs in \$ $t^{-1}_{CO_2eq}$
  b) P.C is the production cost in \$ t^{-1} of product
  c) GWP is the global warming potential in t_{CO_2eq} t^{-1} of product
  d) alt are the alternative routes of production
  e) BAU is the business as usual route of production

- The avoidance costs follow an inverse relation to the natural gas price. As natural gas prices increase, we spend less money for every ton of carbon dioxide emissions avoided.
- Negative avoidance costs signify that the cost of production of BAU ammonia or BAU methanol are higher than the alternative route and we save money for every ton of carbon dioxide emissions avoided.
- Since September 2021, we save around 56$ to 420$ per ton of carbon dioxide emissions avoided by using the biomass route for the production of ammonia. Similarly, from November 2020 for methanol via biomass, savings oscillate from 0.26$ to around 874$ per ton of carbon dioxide avoided.

Table 1 Average monthly prices of natural gas and electricity

Month	Natural gas [$ t^{-1}]	Electricity [$ MWh^{-1}]
Jan '19	326.18	49.41
Feb '19	268.54	42.81
Mar '19	231.45	30.61
Apr '19	219.83	36.97
May '19	193.92	37.83
Jun '19	160.41	32.50
Jul '19	161.75	39.71
Aug '19	164.43	36.82
Sep '19	188.11	35.77
Oct '19	226.09	36.95
Nov '19	230.11	41.00
Dec '19	206.43	31.97
Jan '20	162.20	34.98
Feb '20	130.02	21.91
Mar '20	121.53	22.52
Apr '20	94.73	17.05
May '20	70.60	17.60
Jun '20	78.19	26.23
Jul '20	80.43	30.07
Aug '20	127.79	34.89
Sep '20	176.49	43.68
Oct '20	218.49	33.96
Nov '20	216.26	38.80
Dec '20	261.84	43.55
Jan '21	324.84	52.79
Feb '21	275.24	48.71
Mar '21	273.90	47.18
Apr '21	319.48	53.66

May '21	398.12	53.33
Jun '21	460.22	74.13
Jul '21	558.97	81.29
Aug '21	689.44	82.81
Sep '21	1020.53	128.34
Oct '21	1387.37	139.54
Nov '21	1234.11	176.25
Dec '21	1699.25	220.96
Jan '22	1262.71	167.87
Feb '22	1216.69	128.78
Mar '22	1894.06	251.76
Apr '22	1438.75	166.00
May '22	1303.37	177.51
Jun '22	1499.52	218.18
Jul '22	2293.52	315.26
Aug '22	3129.51	469.35
Sep '22	2473.28	346.12

Table 2 Average CEPCI indexes

Year	CEPCI
2019	607.5
2020	596.2
2021	708.0
2022 (average till July)	816.1

Table 3 Cost parameters used in economic calculations. All costs are presented in $\$_{2019}$.

Component	Average	Low	High
Water [$ t^{-1}]	0.18	-	-
Cooling water [$ GJ^{-1}]	0.38	-	-
CO_2 capture and storage [$ t^{-1}]	15.84	-	-
CO_2 emissions [$ t^{-1}]	30.00	-	-
Nitrogen [$ t^{-1}]	91.06	-	-
Carbon dioxide – DAC [$ t^{-1}] *	137.35	67.85	206.85

*These prices do not include the cost of natural gas required for the capture of carbon dioxide. 1 ton of carbon dioxide requires 0.11 ton of natural gas.

Table 4 Levelized cost of electricity in $ MWh^{-1} and capacity factors (CF)

	2019				2020			
	Average	**Low**	**High**	**CF**	**Average**	**Low**	**High**	**CF**
Wind - on shore	67	37	96	35.6	45	35	65	36.0
Solar PV	68	52	190	17.5	57	39	163	16.1
Nuclear	87	63	112	95.0	69	42	102	95.0

	2021				2022			
	Average	**Low**	**High**	**CF**	**Average**	**Low**	**High**	**CF**
Wind - on shore	42	26	59	39.0	48	30	68	39.0
Solar PV	48	29	120	17.2	55	33	138	17.2
Nuclear	82	50	121	95.0	94	57	140	95.0

Table 5 Levelized cost of hydrogen in $ t^{-1}

	2019			2020		
	Average	**Low**	**High**	**Average**	**Low**	**High**
Wind - on shore	6296	4761	7780	5138	4626	6162
Solar PV	8444	7625	14688	8278	7357	13703
Nuclear	6316	5053	7579	5138	4626	6162
Biomass with CCS	2270	-	-	2228	-	-

	2021			2022		
	Average	**Low**	**High**	**Average**	**Low**	**High**
Wind - on shore	4894	4075	5764	5641	4697	6644
Solar PV	7507	6534	11192	8653	7532	12900
Nuclear	6102	5494	7317	7033	6333	8434
Biomass with CCS	2646	-	-	3049	-	-

Table 6 Global warming potentials in kg_{CO2eq} kg^{-1} ammonia for all scenarios

Process	GWP
BAU	2.180
Wind	0.712
Solar	1.140
Nuclear	0.126
Biomass	-1.140
H_2 SMR + CCS	1.220

Table 7 Global warming potentials in kg_{CO2eq} kg^{-1} methanol for all scenarios

Process	GWP
BAU	0.658
Wind	-0.556
Solar	0.226
Nuclear	-0.686
Biomass	-1.790
H_2 SMR + CCS	0.169

Table 8 BAU ammonia vs green ammonia. All prices are presented in $ t^{-1}. The values highlighted in green are the cases where the alternative route for ammonia production is cheaper than or similar to (not more than 5%) BAU ammonia.

	Naturalgas	BAU	Wind	Solar	Nudear	Biomass	SMR+CCS
Jan '19	326.18	470.06	1337.28	1715.73	1340.86	627.87	636.94
Feb '19	268.54	433.53	1334.18	1712.63	1337.76	624.77	593.64
Mar '19	231.45	402.59	1328.45	1706.90	1332.03	619.04	562.04
Apr '19	219.83	402.42	1331.44	1709.89	1335.02	622.03	556.93
May '19	193.92	389.57	1331.84	1710.29	1335.42	622.43	539.25
Jun '19	160.41	366.93	1329.34	1707.79	1332.91	619.93	513.38
Jul '19	161.75	374.38	1332.72	1711.17	1336.30	623.32	517.70
Aug '19	164.43	373.09	1331.37	1709.82	1334.94	621.96	518.21
Sep '19	188.11	384.58	1330.87	1709.32	1334.45	621.46	534.23
Oct '19	226.09	405.69	1331.43	1709.88	1335.01	622.02	561.28
Nov '19	230.11	411.60	1333.33	1711.78	1336.91	623.92	565.99
Dec '19	206.43	390.68	1329.09	1707.54	1332.67	619.68	545.23
Jan '20	162.20	365.51	1122.68	1675.93	1122.68	609.85	508.60
Feb '20	130.02	336.34	1116.54	1669.79	1116.54	603.70	480.02
Mar '20	121.53	332.43	1116.83	1670.08	1116.83	603.99	474.39
Apr '20	94.73	313.20	1114.26	1667.51	1114.26	601.42	453.12
May '20	70.60	301.00	1114.52	1667.76	1114.52	601.68	436.55
Jun '20	78.19	313.08	1118.57	1671.82	1118.57	605.73	445.90
Jul '20	80.43	317.84	1120.38	1673.62	1120.38	607.54	449.27
Aug '20	127.79	347.30	1122.64	1675.89	1122.64	609.80	484.56
Sep '20	176.49	381.17	1126.77	1680.02	1126.77	613.93	522.66
Oct '20	218.49	394.21	1122.20	1675.45	1122.20	609.37	547.39
Nov '20	216.26	397.56	1124.48	1677.73	1124.48	611.64	548.11
Dec '20	261.84	426.00	1126.71	1679.96	1126.71	613.87	582.13
Jan '21	324.84	514.22	1125.63	1586.07	1338.50	729.49	701.49
Feb '21	275.24	484.28	1123.71	1584.15	1336.58	727.57	664.98
Mar '21	273.90	482.14	1122.99	1583.43	1335.86	726.85	663.33
Apr '21	319.48	512.21	1126.04	1586.48	1338.90	729.90	698.16
May '21	398.12	553.32	1125.88	1586.32	1338.75	729.74	752.85
Jun '21	460.22	605.49	1135.65	1596.09	1348.52	739.52	805.95
Jul '21	558.97	664.20	1139.02	1599.46	1351.89	742.88	878.18
Aug '21	689.44	734.34	1139.73	1600.17	1352.60	743.60	969.90
Sep '21	1020.53	951.32	1161.13	1621.57	1374.00	764.99	1222.22
Oct '21	1387.37	1155.02	1166.39	1626.83	1379.26	770.25	1483.35
Nov '21	1234.11	1108.63	1183.64	1644.08	1396.51	787.50	1393.70
Dec '21	1699.25	1395.45	1204.65	1665.09	1417.52	808.51	1739.14
Jan '22	1262.71	1160.71	1347.78	1878.52	1593.15	891.16	1478.44
Feb '22	1216.69	1099.90	1329.41	1860.15	1574.78	872.79	1427.97
Mar '22	1894.06	1571.73	1387.20	1917.94	1632.57	930.58	1958.21
Apr '22	1438.75	1251.69	1346.90	1877.64	1592.27	890.28	1600.35
May '22	1303.37	1191.14	1352.31	1883.05	1597.68	895.69	1511.33
Jun '22	1499.52	1332.50	1371.42	1902.16	1616.79	914.80	1667.25
Jul '22	2293.52	1841.52	1417.04	1947.78	1662.41	960.42	2266.66
Aug '22	3129.51	2426.00	1489.44	2020.19	1734.81	1032.82	2922.16
Sep '22	2473.28	1965.08	1431.54	1962.28	1676.91	974.92	2406.54

Table 9 BAU methanol vs green methanol. All prices are presented in $ t^{-1}. The values highlighted in green are the cases where the alternative route for methanol production is cheaper than or similar to (not more than 5%) BAU methanol.

	Naturalgas	BAU	Wind	Solar	Nudear	Biomass	SMR+CCS
Jan '19	326.18	409.35	1456.12	1868.48	1460.02	332.25	693.03
Feb '19	268.54	366.31	1445.69	1858.04	1449.59	331.59	638.79
Mar '19	231.45	332.99	1437.63	1849.99	1441.53	330.36	602.55
Apr '19	219.83	329.76	1436.82	1849.18	1440.72	331.00	592.92
May '19	193.92	313.12	1432.78	1845.14	1436.68	331.09	569.18
Jun '19	160.41	287.04	1426.46	1838.82	1430.36	330.55	537.39
Jul '19	161.75	293.04	1427.90	1840.25	1431.79	331.28	539.84
Aug '19	164.43	292.78	1427.84	1840.20	1431.74	330.99	541.83
Sep '19	188.11	307.80	1431.49	1843.85	1435.39	330.88	563.47
Oct '19	226.09	333.91	1437.83	1850.19	1441.73	331.00	598.68
Nov '19	230.11	339.45	1439.17	1851.53	1443.07	331.41	603.07
Dec '19	206.43	317.30	1433.81	1846.17	1437.71	330.50	579.72
Jan '20	162.20	287.06	1201.41	1804.23	1201.41	324.71	532.31
Feb '20	130.02	256.40	1194.00	1796.82	1194.00	323.40	500.45
Mar '20	121.53	251.18	1192.73	1795.55	1192.73	323.46	492.73
Apr '20	94.73	229.46	1187.47	1790.29	1187.47	322.91	467.10
May '20	70.60	213.79	1183.67	1786.49	1183.67	322.97	444.95
Jun '20	78.19	224.96	1186.35	1789.17	1186.35	323.84	453.41
Jul '20	80.43	229.16	1187.36	1790.18	1187.36	324.22	456.12
Aug '20	127.79	264.10	1195.84	1798.66	1195.84	324.71	500.59
Sep '20	176.49	302.74	1205.20	1808.02	1205.20	325.59	546.96
Oct '20	218.49	323.81	1210.34	1813.16	1210.34	324.61	584.03
Nov '20	216.26	325.75	1210.80	1813.62	1210.80	325.10	582.79
Dec '20	261.84	359.45	1218.97	1821.79	1218.97	325.58	625.60
Jan '21	324.84	436.87	1218.04	1719.74	1449.98	386.74	755.90
Feb '21	275.24	400.97	1209.33	1711.03	1441.28	386.33	709.50
Mar '21	273.90	398.99	1208.86	1710.55	1440.80	386.17	708.01
Apr '21	319.48	433.91	1217.32	1719.02	1449.26	386.82	751.11
May '21	398.12	486.02	1229.98	1731.68	1461.92	386.79	823.53
Jun '21	460.22	542.08	1243.54	1745.23	1475.48	388.88	884.29
Jul '21	558.97	612.88	1260.71	1762.41	1492.66	389.60	976.51
Aug '21	689.44	700.80	1282.07	1783.76	1514.01	389.75	1097.01
Sep '21	1020.53	953.40	1343.29	1844.99	1575.24	394.33	1409.86
Oct '21	1387.37	1205.49	1404.50	1906.19	1636.44	395.45	1749.86
Nov '21	1234.11	1129.47	1385.92	1887.62	1617.86	399.14	1614.81
Dec '21	1699.25	1470.71	1468.68	1970.38	1700.63	403.64	2051.06
Jan '22	1262.71	1170.55	1565.72	2144.02	1833.08	456.54	1708.09
Feb '22	1216.69	1112.25	1551.68	2129.97	1819.03	452.61	1659.06
Mar '22	1894.06	1650.16	1681.98	2260.28	1949.34	464.97	2304.16
Apr '22	1438.75	1286.41	1593.87	2172.17	1861.23	456.35	1870.03
May '22	1303.37	1204.44	1573.93	2152.22	1841.28	457.51	1747.19
Jun '22	1499.52	1363.79	1612.51	2190.81	1879.87	461.59	1934.85
Jul '22	2293.52	1960.98	1757.30	2335.60	2024.65	471.35	2683.05
Aug '22	3129.51	2626.49	1918.51	2496.80	2185.86	486.83	3479.60
Sep '22	2473.28	2102.48	1791.58	2369.88	2058.94	474.45	2853.95